

MAT1001 Midterm Practice

1. [5] True or False (in general)? No explanation is required.

(i) Suppose that $g(x) \leq f(x) \leq h(x)$ for all $x > 0$, and the limits of g and h both exist as x approaches 3. Then $\lim_{x \rightarrow 3} f(x)$ must exist.

$x \rightarrow 3$

False



(ii) The function $f(x) = x + \cos(x)$ is increasing on $(-\infty, \infty)$.

$$f'(x) = 1 - \sin(x)$$

True

(iii) If neither $\lim_{x \rightarrow c} f(x)$ nor $\lim_{x \rightarrow c} g(x)$ exists, then $\lim_{x \rightarrow c} f(x) + g(x)$ does not exist.

$x \rightarrow c$

$x \rightarrow c$

$x \rightarrow c$

False

(iv) If $f(4) = -\sqrt{5}$, $f(8) = \pi$, and f is differentiable everywhere, then the equation $f(x) = 0$ must have at least one solution on the interval $(4, 8)$.

True

(v) If f is continuous on $[a, b]$ then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

Local, not absolute False

(vi) Using the midpoint rule to estimate the area under the graph of $f(x) = 4 - x^2$ and above the x -axis on $[-2, 0]$ will definitely give an overestimate of the true area.

True

1.75 3.75

$$4x - \frac{x^3}{3} \Big|_{-2}^0$$

2. [10] Short questions: no explanation is required.

(i) The function $f(x) = |3x - 2|$ is differentiable over which of the following intervals?

A) $(-4, 4)$

☒ B) $(-5, 1)$

C) $(1, 5)$

D) None of the above

ii) Let $f(x) = |x|$ and $g(x) = f'(x)$, compute $\lim_{x \rightarrow \infty} g(x)$.

$x \rightarrow \infty$

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iii) Suppose that $g(x) = \frac{f(x)}{h(x)}$, where f and h have derivatives of all orders. Express $g'(x)$ in terms of f , h , and their first derivatives.

$$g'(x) = \frac{h(x)f'(x) - f(x)h'(x)}{(h(x))^2}$$

iv) A particle is moving along the y -axis, where its position at time t is given by $y(t) = \sin(4t) - 5\cos(t)$, where t and y are in seconds and meters respectively.

Determine the velocity and acceleration of this particle at $t = \pi$.

$$v(t) = 4\cos(4t) + 5\sin t$$

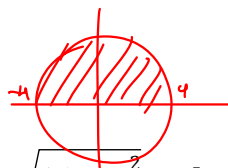
$$a(t) = -16\sin(4t) + 5\cos t$$

$$v(\pi) = 4$$

$$a(\pi) = -5$$

v) Find the function $f(x)$ that satisfies $f'(x) = 2x + 7$ and $f(0) = 5$.

$$x^2 + 7x + 5$$



vi) Find the average value of $f(x) = \sqrt{16 - x^2}$ on $[-4, 4]$. $AV(f) = \frac{\int_{-4}^4 f(x) dx}{4 - (-4)} = \frac{8\pi}{8} = \pi$

vii) Find the norm of the partition $P = \{-2, -1.6, -0.5, 0, 0.8, 1\}$. 1.1

viii) Express the following limit as definite integral: $\lim_{||P|| \rightarrow 0} \sum_{k=1}^n \frac{1}{c_k} \Delta x_k$ $\int \frac{1}{x} dx$

ix) Find the linearization of $f(x) = \cos(x)$ at $x = \frac{\pi}{2}$. $\cos(\frac{\pi}{2}) - \sin(\frac{\pi}{2})(x - \frac{\pi}{2}) = -x + \frac{\pi}{2}$

x) Consider the equation $x^3 + 3x + 1 = 0$, use Newton's method to estimate the solution, starting with $x_0 = 0$, find x_1 and x_2 .

$$x_0 = 0 \rightarrow x_1 = 0 - \frac{1}{3} = -\frac{1}{3}$$

$$x_1 = -\frac{1}{3} \rightarrow x_2 = -\frac{1}{3} - \left(\frac{-\frac{1}{27} + 1}{-\frac{1}{9} + 3} \right) = -\frac{1}{3} + \frac{1}{90} = -\frac{29}{90}$$

3.[5+5+10] Evaluate the following limits. DO NOT USE methods you find in chapters outside of the midterm scope (eg : chapter 7).

(i) $\lim_{x \rightarrow 0^+} \sqrt{x} (1 + \sin^2(\frac{2\pi}{x}))$ Squeeze: $0 \leq \sin^2(\frac{2\pi}{x}) \leq 1$

$$1 \leq \sin^2(\frac{2\pi}{x}) + 1 \leq 2$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} \leq \lim_{x \rightarrow 0^+} \sqrt{x} (\sin^2(\frac{2\pi}{x}) + 1) \leq \lim_{x \rightarrow 0^+} 2\sqrt{x}$$

$$0 \rightarrow \lim_{x \rightarrow 0^+} \sqrt{x} (1 + \sin^2(\frac{2\pi}{x})) = 0$$

$$= \lim_{x \rightarrow \infty} \frac{3x+1}{\sqrt{x^2+1} + \sqrt{x^2-3x}}$$

$$= \lim_{x \rightarrow \infty} \frac{3x+1}{x\sqrt{1+\frac{1}{x^2}} + x\sqrt{1-\frac{3}{x}}}$$

$$= \lim_{x \rightarrow \infty} \frac{3x}{x(1+\frac{1}{x^2}) + x(1-\frac{3}{x})} = \frac{3}{2}$$

(ii) $\lim_{x \rightarrow \infty} \sqrt{x^2+1} - \sqrt{x^2-3x}$

(iii) Suppose that $f(x)$ is continuous $\forall x \in [-1, 1]$ and $f(0) = 3$. Determine the value of the following limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+f(x)\sin(x)} - 1}{x} = \lim_{x \rightarrow 0} \frac{f(x)\sin(x)}{x(1+\sqrt[3]{1+f(x)\sin(x)})^2} = 1$$

4.[5+5] Given the curve defined by the equation $xy + 3x = y^2$: $x \frac{dy}{dx} + y + 3 = 2y \frac{dy}{dx}$

(i) Find the equation of the tangent line to the curve at $(\frac{1}{4}, 1)$. $\frac{y+3}{2y-x} = \frac{dy}{dx}$

(ii) Find the second derivative of y with respect to x at $(0, 1)$.

$$\frac{1}{4}, 1: \frac{dy}{dx} = \frac{4}{7} = \frac{16}{7}$$

$$f'(x) = -2\cos x \sin x - 2\cos x = -2\cos x(\sin x + 1) > 0 \text{ when } x > \frac{\pi}{2}, \frac{3\pi}{2}$$

5.[5+5+5] Let $f(x) = \cos^2(x) - 2\sin(x)$ with domain $D = [0, 2\pi]$.

(i) Find the closed interval(s) where $f(x)$ is strictly increasing or strictly decreasing.

(ii) Find all the critical points of $f(x)$ in the interior of D . $(\frac{\pi}{2}, -2), (\frac{3\pi}{2}, 2)$

(iii) Find the x values of all local maxima and minima of $f(x)$ in D .

$$f'(x) = 3x^2 - 12x + 6 \rightarrow f''(x) = 6x - 12$$

6.[10] Let $f(x) = x^3 - 6x^2 + 6x$ with domain $D = (-\infty, \infty)$, find the intervals where f is concave up, concave down, and find all the inflection points.

Concave down: $x < 2$
Concave up: $x > 2$

7.[10] Let $f(x) = \frac{16x^2 - 300x + 1569}{2x + 60}$ and $g(x) = x + x\sin(\frac{1}{x})$, find the vertical, horizontal, and oblique asymptotes for $f(x)$ and oblique asymptotes only for $g(x)$.

$$y = 8x - 390 + \frac{24969}{2x + 60}$$

Oblique

$$y = x$$

$f(x)$ is increasing $\rightarrow f(-1) < f(1)$ ($c \in [-1, 1]$)

$$\therefore f'(x) < \frac{1}{2} \forall x \rightarrow \frac{f(1) - f(-1)}{1 - (-1)} = f'(c) < \frac{1}{2} \rightarrow f(1) - f(-1) < 1$$

8.[5+5] (i) Suppose that $\forall x \in R, 0 < f'(x) < \frac{1}{2}$, show that $f(-1) < f(1) < 2 + f(-1)$. $f(1) < 1 + f(-1) < 2 + f(-1)$

(ii) Let $f(x) = px^2 + qx + r$ for all $x \in R$, show that f can have at most two real zeros.

$$f'(x) = 2px + q, \quad f''(x) = 2p \neq 0$$

$\rightarrow$ exist one zero

$$f > 2 \text{ zeros} \rightarrow \frac{A}{3} \quad 2\pi\left(\frac{A}{6\pi}\right)\left(\frac{\sqrt{A}}{\sqrt{6\pi}}\right)$$

$$A = 2\pi r^2 + 2\pi r h = 2\pi r(r+h)$$

9.[10] You have a certain area A of material to make a cylindrical can of radius r and height h .

The material of area A includes the top, the bottom and the side of the cylinder. Find the values of r and h so that the volume of the cylinder is maximized.

Find max $\pi r^2 h$

$$V = \pi r^2 \left(\frac{A}{2\pi r} - r\right)$$

$$V = \frac{Ar}{2} - \pi r^3$$

$$0 = \frac{dV}{dr} = \frac{A}{2} - 3\pi r^2$$

$$r = \frac{\sqrt{A}}{\sqrt{6\pi}} \rightarrow h = \frac{\sqrt{2A}}{\sqrt{3\pi}}$$

10. [2+8]

$$2\sqrt{2}(1-0) < \int_0^1 \sqrt{x+8} dx < 3(1-0)$$

$$2\sqrt{2} < \sqrt{x+8} < 3$$

(i) Show that the value of $\int_0^1 \sqrt{x+8} dx$ lies between $2\sqrt{2}$ and 3.

(ii) Let $f(x) = x + x^2$, find a formula for the Riemann sum obtained by dividing the interval $[0, 1]$ into n equal subintervals and using the right-hand endpoint for each c_k . Then take a limit of

these sums as $n \rightarrow \infty$ to calculate the area under the curve over $[0, 1]$.

$$c_i = \frac{i}{n}$$

$$\Delta x_i = \frac{1}{n}$$

$$\sum_{i=1}^n \left(\frac{i}{n} + \left(\frac{i}{n}\right)^2\right) \left(\frac{1}{n}\right) = \sum_{i=1}^n \left(\frac{i}{n^2} + \frac{i^2}{n^3}\right) = \frac{n(n+1)}{2n^2} + \frac{n(n+1)(2n+1)}{6n^3}$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k = \lim_{n \rightarrow \infty} \left(\frac{n(n+1)}{2n^2} + \frac{n(n+1)(2n+1)}{6n^3} \right) = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

11. [Bonus]

(i) Let $g(x) = \int_{2022}^x (e^{t^{2023}} - \sin^{2024}(t)) dt$, find $g'(0)$.

$$g'(0) = e^{0^{2023}} - \sin^{2024}(0) \Big|_{t=0} = 1 - 0 = 1$$

(ii) If $f(x)$ is continuous everywhere, find the value of $I = \int_0^a \frac{f(x)}{f(x)+f(a-x)} dx \Rightarrow \frac{a}{2}$

$$a - I = \int_0^a \frac{f(a-x)}{f(x)+f(a-x)} dx = - \int_a^0 \frac{f(u)}{f(a-u)+f(u)} du$$

$$u = a-x \quad -du = dx \quad = \int_0^a \frac{f(x)}{f(a-x)+f(x)} dx = I$$

$$a = 2I$$

End of Paper

Comments and Additional Notes :

- 1) All questions here are either easy or slightly-intermediate. Good news is, although easy, pretty much all the midterm-emphasized topics are covered in this whole practice.
- 2) For topics such as Newton's Method and all those in 5.1-5.3, simply read the book (understand the graphs).
- 3) Proofs are less likely to show up in ur exam, so read them if you are curious but otherwise you're good anyway.
- 4) Just for additional reference, for some of you who want to know how much grade is "good" enough (NOTE!! This is purely estimate and completely subjective based on past observations)
 - i) Good : $\max_{score} - 20 \leq grade \leq \max_{score}$
 - ii) Average : $\max_{score} - 40 \leq grade \leq \max_{score} - 20$
 - iii) Try your best not to get below average lol, finals a lot harder so better do this midterm well so you don't need to plan on doing an academic "comeback" on the finals.
- 5) Lastly, for those of you whose midterm score lies in the range (i) above, you may inform me through Wechat or whatever if you are interested in joining some team-based competitions (involving math/programming; date still tentative).
- 6) Good luck and all the best.